

YEAR 10 BIOLOGICAL SCIENCE
BIOLOGICAL CHEMISTRY
BLOOD ALCOHOL LEVEL

AIM To determine which “blood samples” contain an alcohol level above the legal maximum.

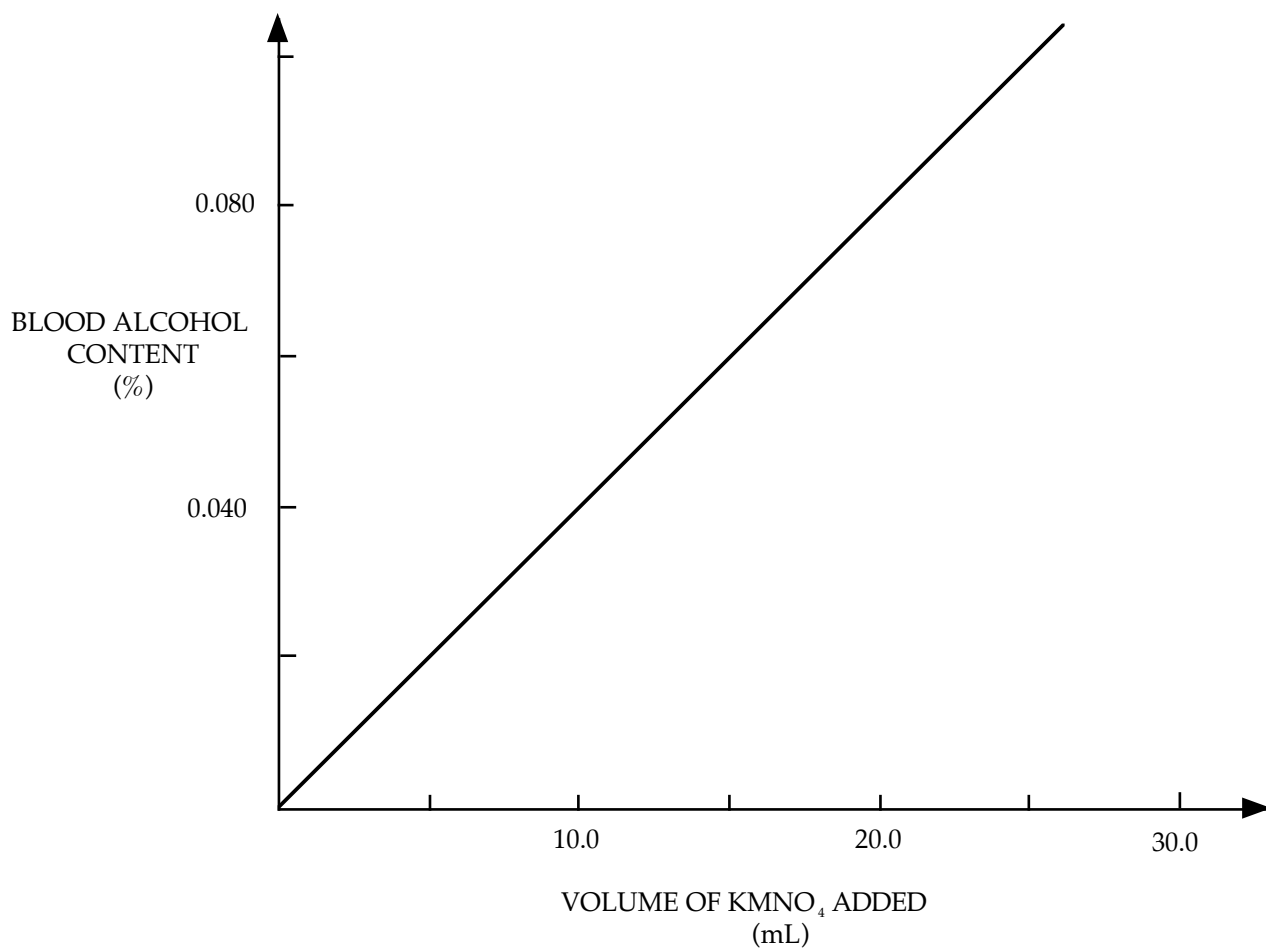
- METHOD**
1. Fill a burette with $0.0200 \text{ mol L}^{-1}$ potassium permanganate (KMnO_4) solution.
 2. Pipette 20.00 mL of the “blood sample” A into a conical flask and add about 20.00 mL of $2.0 \text{ mol L}^{-1} \text{ H}_2\text{SO}_4$.
 3. Warm the conical flask to about 80°C .
 4. Record the initial level of KMnO_4 in the burette.
 5. Run the KMnO_4 solution into the conical flask slowly while swirling it. Continue until the solution turns permanently pink. Turn off the burette tap and record the final volume of KMnO_4 in the burette.
 6. Calculate the amount of KMnO_4 added by subtracting the initial volume from the final volume.
 7. Use the graph provided to determine the alcohol content of the sample.
 8. Repeat the procedure for samples B and C.

RESULTS

VOLUME OF KMnO_4 ADDED (mL)	BLOOD SAMPLES		
	A	B	C
FINAL READING			
INITIAL READING			
TITRATION VOLUME			

ALCOHOL READINGS

Sample A	_____
Sample B	_____
Sample C	_____



CONCLUSION